# Measuring resistance versus temperature in metal and semiconducting samples

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January 2022

**Abstract**

The purpose of the experiment is to analyse temperature dependance of resistivity of two metals (copper and nickel) and a semiconductor (germanium). The two metals does not perfeclty follow the linear behaviour expected by Wiedemann–Franz law: the temperature coefficient is temperature dependent. By analysing germanium resistivity, we can estimate the energy gap $E_g$ fitting in the intrinsic region, at high temperatures. The estimated value of 0.63 eV is coherent with other experimental data. In the extrinsic region the polynomial temperature dependance of mobility can be extimated as $T^{-1.41}$. Two-wire and four-wire resistance measurement of some unknown metal samples are performed and the composition of the samples is guessed.

## Introduction

Electrical resistivity vs. Temperature profiles are able to verify microscopical models of conductivity in metals and semiconductors. In general, resistivity $\rho$ is inversely proportional to the carrier concentration $n$ and mobility $\mu$. Discrepancies between classical and quantum model are found in the evaluation of mobility: in according to Drude model $\mu \propto T^{-1/2}$, while a more precise quantum model predicts $\mu \propto T^{-1}$. Classically, electrical resistance is the result of scattering between electrons accelerating in an electric field and fixed ions. Actually, in a perfect lattice there is no scattering: the finite resistivity at $T = 0$ K is due to impurities and the linear profile with temperature is due to lattice vibrations, whose energy is proportional to thermal energy. In a semiconductor the carrier concentration depends on temperature and doping. In a n-doped semiconductor at low temperatures, the fraction of thermally generated carriers is negligible: the carrier density is constant and equal to the dopant concentration $N_d$. For high temperatures, thermally generated carriers $n_i(T) \propto \exp(-E_g(T)/2k_BT) \gg N_d$. Theese two limits define respectively the extrinsic and intrinsic regions. Mobility decreases as $\mu \propto T^{-\alpha}$, where $\alpha$ depends on chemical composition. Resistance measurement are performed generating and measuring current in a loop and measuring voltage via another high impedance loop. Theese measurements are generally affected by two type of systematic errors: in a simple two-wire measurement, wire's resistance adds in series to sample's resistance, because amperometer and voltmeter's electrodes are not separated. This effect cannot be neglected if sample's resistance is low. In a four-wire measurement two different pairs of electrodes are used: wire's resistance of amperometer's loop is no longer inside the voltmeter's loop and a trascurable current passes through voltmeter's wire resistance. In a semiconductor, metal-semiconductor

contact resistance cannot be neglected, therefore four indipendent contancts must be realized in order to bring contact resistance inside the high impedance voltmeter's branch.

## Materials and Design

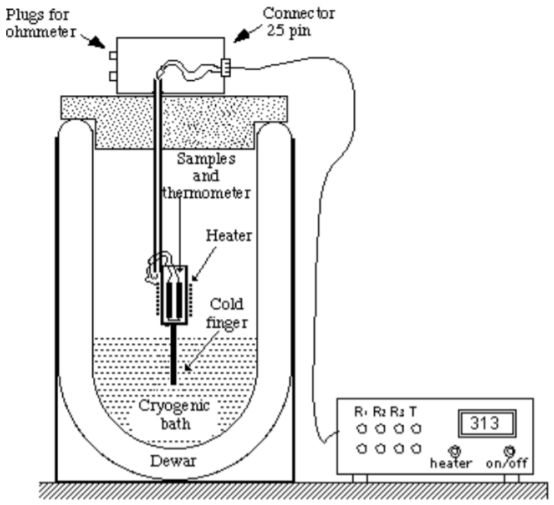
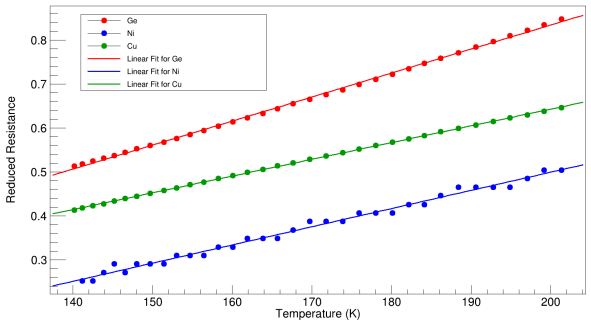



*Figure 1: Experimental apparatus. The ohmmeter at the top left allow two-wire measurement, the control unit at the bottom right allow four-wire measurement.*

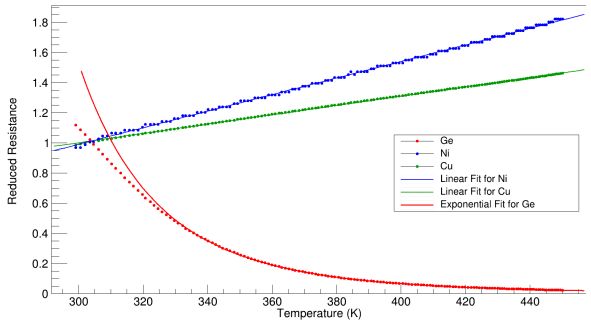
The inital setup consists in a dewar, containing a sample holder in thermal contact with a liquid nitrogen bath via a metallic cold finger. The holder is surrounded with a 50 Ω heater. This apparatus allows measurements in a 100-460 K temperature range. The samples taken in analysis are composed of germanium ($R_1$), copper ($R_2$) and nickel ($R_3$). A forward-biased PN juction measures the temperature $T$. A constant current passes through samples and thermometer; voltage measurements are taken with a different pair of contacts, realizing a four-wire measurement. The metallic samples are obtained from 50 $\mu$m wires; the germanium bar is $3 \times 5$ mm and 1 mm thick, with four wires soldered at the corners, realizing a four-contact setup. In order to display the systematic difference between a two-wire and a four-wire measurement, resistance of four wires of unknown metallic materials are measured.

## Results and Discussion

### Resistance versus Temperature




(a)




(b)

*Figure 2: Low (a) and high (b) temperatures regime of resistance of the three samples, fitted with linear functions (a), and with exponential function for germanium in (b).*

Resistance values can be normalized measuring $R_0$, definded as the resistance at room temperature ($T = 300$ K), and plotting the ratio $R/R_0$. This ratio is also equal to the ratio of resistivity $\rho/\rho_0$. Fig. (2a) shows resistance behaviour in the low temperatures regime. Metal's resistance grows approximately linearly in both regimes, while germanium resistance $R_1$ shows two different behaviours. At low temperatures ($T < 200$) K, $R_1$ depends only on mobility, which shows a $T^{-\alpha}$ behaviour, with $\alpha = 1.41 \pm 0.03$.

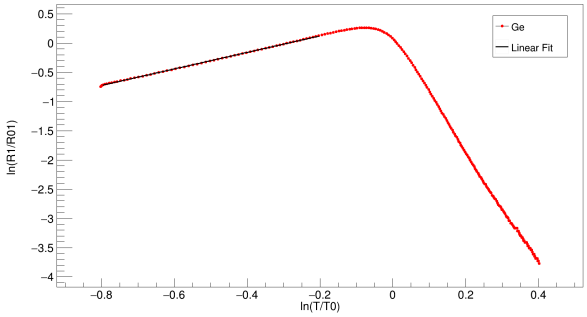
This coefficient varies with the range of fitting, therefore the error is taken as the disperion of the value for ranges within the region taken in analysis. In the high temperatures regime ($T > 350$ K), $R_1$ depends almost exclusively on intrinsic carrier concentration, which has an exponential behaviour. From the fit in Fig. (3), the energy gap $E_g$ can be estimated as 0.63 eV. This analysis doesn't take into account polynomial dependance of mobility and energy gap; therefore the estimated value of $E_g$ correspond to a temperature range from 300 K to 450 K, in which the literature value of $E_g$ ranges from 0.66 eV to 0.60 eV [1] [2]. Linear coefficients of nickel and copper are, respectively, $\beta_2 = (3.4 \pm 0.4)\ K^{-1}$ and $\beta_3 = (4.8 \pm 0.7)\ K^{-1}$. Theese two values are obtained averaging the two coefficient obtained from fitting at low and high temperatures. At 293 K, literature values are $\beta_2 = 4.0\ K^{-1}$ [3] and $\beta_3 = 6.4\ K^{-1}$ [4]. This deviation is due to the fact that resistance behaviour of metals is not perfectly linear, therefore averaging the coefficient in different temperature regimes is not ideal.

## Resistance of unknown samples

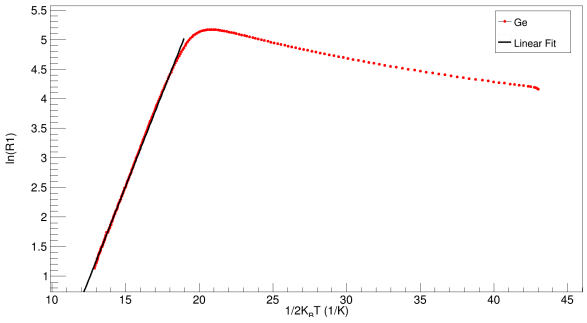
Two-wire and four-wire measurements are performed varying the distance from voltmeter's contacts $d$, and then averaging the calculated values of the resistivity $\rho$. From the table below, it is clear that there is a substantial difference in measuring with the two different tecniques. Comparing the results of four-wire measures with literature and taking into account also the appearance of the wires, we estimate that:

| | Two-wire $\rho$ $[\times 10^{-8} \Omega m]$ | Four-wire $\rho$ $[\times 10^{-8} \Omega m]$ |
|---|---|---|
| A | $208 \pm 7$ | $3 \pm 2$ |
| B | $173 \pm 5$ | $117 \pm 4$ |
| C | $57 \pm 1$ | $8.1 \pm 0.6$ |
| D | $9 \pm 1$ | $2.0 \pm 0.1$ |

- Sample A is aluminum, the wire is slightly oxidized and $\rho = 2.65 \times 10^{-8}\ \Omega m$ [5];
- Sample B is nichrome, $\rho = 100 \times 10^{-8}\ \Omega m$ [5];




(a)




(b)

*Figure 3: Linear fit of germanium resistance. On the axis is plotted the logarithm of resistance and temperature (a) in order to remark the monomial behavior of resistivity at low temperatures. In (b) on the axis is plotted the logarithm of resistance and the inverse of temperature in order to estimate the energy gap value in the fit temperature range.*

- Sample C is tungsten, the wire is more rigid and darker than the other wires, $\rho = 5.6 \times 10^{-8}\ \Omega m$ [5];
- Sample D is copper, the wire's color is red-orange, $\rho = 1.68 \times 10^{-8}\ \Omega m$ [3].

# Conclusions

In the Resistance vs. Temperature section, estimated Resistivity coefficient of copper and nickel are far from literature results: even fitting in a narrow range centered at 293 K (temperature at which the correct values are given), the deviation is $> 10\%$. For germanium, $E_g$ and mobility dependance with temperature is coherent with literature data. In the wire resistivity measurement section, due to the high error on resistivity and high deviation from literature data, the identification can be incorrect.

# References


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